

Depletion-Gated Reorientation: A Minimal Mechanism for Efficient Spatial Foraging

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Preprint — March 2026

Abstract

We study the problem of efficient foraging in patch-depleting spatial environments, where an agent must balance local exploitation with timely abandonment of depleted patches. We propose the **Depletion-Gated Pivot (DGP)**, a minimal GRU-based architecture combining exploration (IEC: full memory updates, high entropy) with a depletion-gated reorientation event: when local viability falls below a threshold, the agent executes a brief high-entropy action selection step — with optional partial memory reset — that disrupts exploitation habits and promotes spatial relocation. We evaluate DGP against ablations and a capacity-matched PPO-GRU baseline across three spatial environments with varied depletion dynamics. DGP consistently outperforms IEC-only exploration ($p < 0.05$ across all environments) and outperforms the PPO-GRU baseline ($p < 0.03$ in all environments), while a richer staged variant (DGP + EIL drift phase) is substantially worse than DGP in all environments, confirming the current EIL-style drift phase is harmful overhead in this environment family. DGP fails in a non-spatial bandit setting where pivot is not coupled to physical relocation, establishing clear scope. Ablations show that performance gains are robust to the strength of the internal state reset (including no reset at $\alpha=1.0$), implicating high-entropy reorientation as the primary driver.

To identify the key mechanism, we run a pre-registered control comparing DGP against **Viability-Conditioned Entropy PPO (VCE-PPO)**: an identical hard step-gate at $\tau=0.25$, but substituting the structured pivot action with increased policy entropy ($\text{entropy_high}=0.10$). VCE-PPO reduces overcommitment (OCI) relative to IEC-only, but fails to improve AUC over IEC-only in any environment ($p > 0.30$). DGP significantly outperforms IEC-only on AUC in all environments ($p < 0.05$) and achieves substantially lower OCI (gap 0.028-0.047, $p < 0.001$ across environments).

This dissociation establishes that the structured spatial pivot action — not entropy modulation — is the key performance-improving mechanism in this environment family.

1. Introduction

Optimal foraging theory (Charnov, 1976) prescribes that agents should leave a patch when its marginal returns fall below the average environment return. Implementing this behaviorally requires detecting depletion, reorienting toward unexplored patches, and building memory of patch quality. In artificial agents, these requirements translate into architectural choices: how should memory be updated, when should exploration switch to exploitation, and how should depletion be detected?

We address these questions through a controlled falsification approach. We implement a family of GRU-based architectures with varying structural complexity and evaluate them in the Hard Patchy Niche (HPN) environment — a 7×7 grid of resource patches with explicit depletion and recovery dynamics. Our primary question is: **what is the minimal architectural mechanism that consistently outperforms unstructured exploration in this environment family?**

Our contributions are: 1. **Depletion-Gated Pivot (DGP)**: A minimal architecture (IEC exploration + depletion gate + high-entropy pivot) that consistently outperforms IEC-only exploration across spatial environments with varied depletion dynamics ($p < 0.05$ in all three environments). 2. **Falsification of richer variants**: A staged architecture adding an Exploitation/Inhibition Loop (EIL drift phase) substantially underperforms DGP in all environments (e.g., AUC 0.189 vs 0.209 on HPN), confirming the current EIL-style drift phase is harmful overhead in this environment family rather than beneficial structure. 3. **Scope characterization**: DGP fails in a non-spatial bandit (MPB), and the scope limitation persists even with forced patch-switching post-pivot, establishing that the mechanism requires spatial structure where the pivot is coupled to physical relocation. 4. **Mechanism attribution via reset sensitivity**: A sensitivity analysis over reset strength ($\alpha \in \{0.0, 0.1, 0.5, 1.0\}$) shows the pivot benefit is robust including $\alpha=1.0$ (no memory wipe), implicating high-entropy reorientation as the primary driver. 5. **VCE-PPO control (pre-registered)**: A Viability-Conditioned Entropy PPO agent using the same hard viability gate but substituting the pivot action with entropy

modulation fails to outperform IEC-only on AUC ($p > 0.30$ in all environments), while DGP significantly beats IEC-only ($p < 0.05$). The critical distinction is not the presence of a viability gate: both DGP and VCE-PPO use an identical hard step-gate at the same threshold. The difference is that DGP couples the gate to a structured spatial pivot action, whereas VCE-PPO couples the gate only to increased policy entropy. This dissociation identifies the structured pivot action as the key performance-improving mechanism in this environment family.

Context and scope. This work was originally motivated by a broader theoretical context concerning lifecycle structure in adaptive systems (Ollar, 2026a; 2026b). The present paper does not test that broader framework; it tests one narrow, falsifiable mechanism-level prediction derived from that line of thinking: that a depletion-gated reorientation step can improve performance beyond entropy modulation alone in spatial foraging settings. All empirical claims rest exclusively on the experiments reported here.

2. Background

Optimal foraging theory. The Marginal Value Theorem (Charnov, 1976) provides the normative solution to the patch-leaving problem: an agent should depart when the instantaneous return rate equals the long-run average environment return. This requires detecting diminishing local returns and coordinating departure with reorientation. Extensions to stochastic (Stephens & Krebs, 1986) and spatially explicit (Olsson & Holmgren, 1999) settings have reinforced the generality of patch-leaving rules across foraging contexts.

Memory and recurrent agents. Recurrent neural networks, particularly LSTMs (Hochreiter & Schmidhuber, 1997) and GRUs (Cho et al., 2014), have been applied to partially observable RL tasks requiring integration over history. Memory-augmented agents (Graves et al., 2016; Ritter et al., 2020) can in principle represent patch quality over episodes. However, how recurrent memory should interact with depletion detection is not prescribed by architecture alone — it is the design question this paper addresses.

Exploration and exploitation in RL. Classical exploration strategies (ϵ -greedy, UCB, Thompson sampling) operate on stationary reward distributions. Restless bandits (Whittle, 1988; Tekin & Liu, 2012) model

non-stationary arms with known dynamics. Patch foraging is a spatially structured restless bandit where the agent’s own actions deplete the active arm and physical movement is required to switch. Meta-RL approaches (Wang et al., 2016; Duan et al., 2016) have explored adapting exploration strategies within episodes, but without targeted patch-leaving mechanisms.

Depletion-aware foraging in RL. Prior work on resource-depleting environments in RL has largely focused on reward shaping or explicit tabular patch tracking (Constantino & Daw, 2015). Our approach differs in targeting the minimal architectural change — a mechanically scheduled gate — that induces efficient patch-leaving without requiring tabular state or learned depletion models.

3. Problem and Environments

3.1 Hard Patchy Niche (HPN)

A 7×7 grid where each cell holds richness $r \in [0, 1]$, initialized from $\text{Exp}(0.7)$ clipped to $[0, 1]$. Actions: EXPLOIT (harvest richness, multiply by δ), MOVE (4-directional). Richness recovers at rate ρ per step for cells not visited in the last 50 steps.

Standard HPN ($\delta = 0.60$, $\rho = 0.03$): balanced depletion/recovery. **HPN-SlowRecovery** ($\delta = 0.60$, $\rho = 0.01$): long-term depletion stress. **HPN-SteepDecay** ($\delta = 0.40$, $\rho = 0.03$): rapid local collapse.

Observation: $[x/6, y/6, r_{\text{local}}, \bar{r}_{\text{neighbours}}, t_{\text{since}}/50, \bar{r}_{\text{global}}] \in \mathbb{R}^6$.

V_t (viability signal): $V_t = r_{\text{local}}$ (obs[2]). Gate fires when $V_t < \tau = 0.25$.

Evaluation: Normalized AUC over 500 training episodes (400 steps each). Each agent trained from scratch per seed; 10 seeds reported.

3.2 Multi-Patch Bandit (MPB)

10 independent patches with depletion dynamics ($\delta = 0.70$, $\rho = 0.02$). Agent selects HARVEST or SWITCH-to-patch actions. No spatial structure — pivot is a memory-only operation. Used to characterize scope of DGP.

3.3 Static Niche World (SNW)

Identical to HPN but richness never depletes ($V_t = 1.0$ constant). Used as sanity check: DGP should not trigger in static environments.

4. Method: Depletion-Gated Pivot (DGP)

4.1 Architecture

DGP is a GRU-based agent with a mechanical scheduler:

- **IEC mode** (default): Full GRU update at temperature $T_{IEC} = 2.0$ (high entropy, exploration).
- **PIVOT** (depletion-triggered): When $V_t < \tau$, execute one pivot step consisting of two components:
 1. **State mixing** (optional): $h \leftarrow \alpha h + (1 - \alpha)\epsilon$, $\epsilon \sim \mathcal{U}(0, I)$, canonical $\alpha = 0.5$. This partially disrupts accumulated exploitation patterns. Sensitivity to α is reported in Appendix B.
 2. **High-entropy reorientation**: Full GRU write at temperature $T_{PIVOT} = 5.0$, promoting action diversity. This is the primary driver of the pivot benefit (see §7.3).

After one PIVOT step, the agent returns to IEC.

The gate fires directly from IEC: IEC $\rightarrow$ check $V_t \rightarrow$ if $V_t < \tau$, PIVOT (one step) $\rightarrow$ return to IEC.

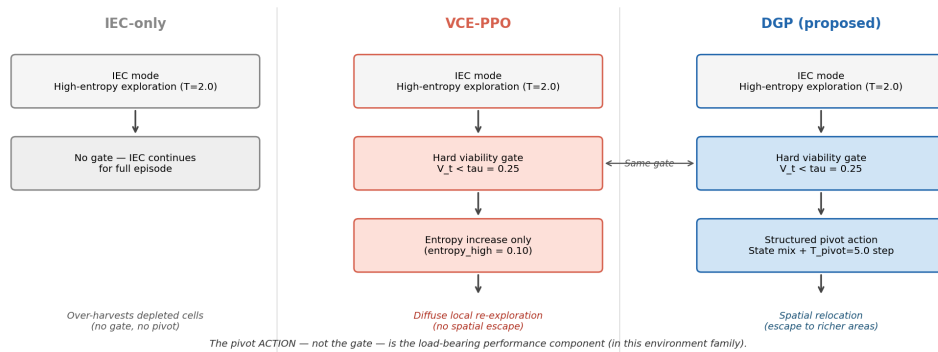


Figure 1

Figure 1. Mechanism comparison. All three agents share the same GRU backbone. IEC-only has no gate and runs exploration continuously. Both

VCE-PPO and DGP use an identical hard step-gate at $\tau=0.25$ — the gate is not the distinguishing factor. VCE-PPO couples the gate to increased policy entropy only; DGP couples the gate to a structured pivot action (partial state reset + high-temperature reorientation step) that drives spatial relocation. The pivot action — not the gate itself — is the load-bearing performance component in this environment family.

4.2 Harness (v1.5b)

V_t is injected from the environment at each step: `agent.set_vt(info['v_t'])`. In HPN, $V_t = r_{\text{local}}$ (raw richness, a low-viability proxy that fires on both depleted and inherently poor cells). In SNW, $V_t = 1.0$ always (gate structurally disabled in static environments).

4.3 Ablations

All ablations use the same GRU backbone with equivalent parameter count:

Agent	Description
DGP (TFP-NoEIL)	IEC + depletion-gated pivot (proposed)
DGP-Full (TFP-Fixed)	IEC + EIL drift phase + depletion-gated pivot
IEC-only	IEC mode only (no pivot, full episode)
EIL-only	EIL drift mode only (no IEC, no pivot)
No-pivot	Alternating IEC+EIL without pivot gate
Random-pivot	IEC+EIL with random (non-gated) pivot
PPO-GRU	PPO with GRU hidden state, capacity-matched

5. Results: HPN Main

5.1 Primary Comparisons

Table 1. Primary results on standard HPN (mean $\pm$ SEM, 10 seeds, 500 training episodes).

Agent	AUC mean $\pm$ SEM
DGP	0.2086 $\pm$ 0.0007
IEC-only	0.2038 $\pm$ 0.0005
EIL-only	0.1789 $\pm$ 0.0005
No-pivot	0.1837 $\pm$ 0.0006
Random-pivot	0.1641 $\pm$ 0.0007

Agent	AUC mean $\pm$ SEM
PPO-GRU	0.2024 $\pm$ 0.0007

DGP outperforms IEC-only ($p=0.0004$), No-pivot ($p=0.0008$), and outperforms PPO-GRU ($p<0.001$).

5.2 Ablation: Why EIL Fails

DGP-Full (0.1889) underperforms IEC-only (0.2038, $p\approx 1.0$ for DGP-Full $>$ IEC-only). The EIL drift phase (7.2% occupancy, mean run 3.3 steps) adds overhead without benefit. Mechanistically, brief EIL bursts do not provide meaningful slow exploitation; EIL-only (pure exploitation drift) also underperforms IEC-only ($0.179 < 0.204$). The combination IEC+EIL+Pivot inherits EIL's costs and adds memory reset overhead.

5.3 Mechanism Summary

DGP triggers ~ 94 pivot events per episode (400 steps; q4b denotes gate-triggered pivot transitions per episode), confirming the gate fires regularly on depleted cells. Each pivot is a single step: IEC mean run length is ~ 3.2 steps before depletion triggers the next pivot. This rapid cycle (exploit $\rightarrow$ deplete $\rightarrow$ pivot $\rightarrow$ move) is the behavioral pattern underlying the AUC gain. Mode occupancy: $\sim 97\%$ IEC, $\sim 3\%$ PIVOT (single-step pivots). Full mechanistic analysis in §7.

6. Generalization Across HPN Variants

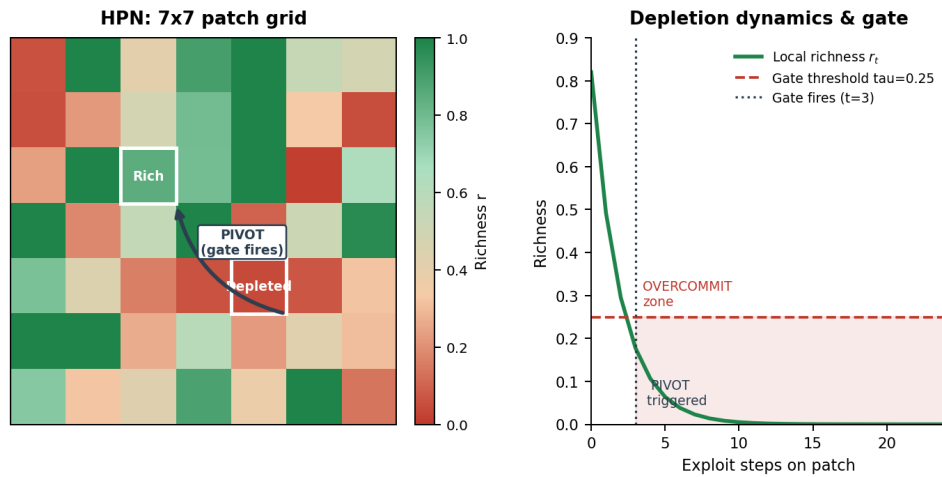


Figure 2

Figure 2. HPN environment schematic. (Left) The 7×7 grid, color-coded by richness (green = rich, red = depleted). The agent exploits a rich patch until local richness falls below $\tau=0.25$ (gate fires), then pivots and relocates. (Right) Depletion dynamics on a single patch under repeated exploitation ($\delta=0.60$): richness declines exponentially until the viability threshold is crossed. DGP responds with a spatial pivot; VCE-PPO responds with entropy increase only. The overcommit zone (shaded) marks steps where IEC-only and VCE-PPO continue exploiting despite sub-threshold richness.

Table 2. AUC across three HPN environments (mean $\pm$ SEM, 10 seeds).

Environment	DGP	VCE-PPO	IEC-only	DGP-Full	PPO-GRU	p (DGP vs IEC)	p (DGP vs PPO-GRU)
HPN standard	0.2086 $\pm$ 0.0007	0.1999 $\pm$ 0.0013	0.2038 $\pm$ 0.0005	0.1889 $\pm$ 0.0019	0.2024 $\pm$ 0.0007	0.001	<0.001
HPN-SlowRecovery	0.1881 $\pm$ 0.0007	0.1811 $\pm$ 0.0009	0.1854 $\pm$ 0.0008	0.1737 $\pm$ 0.0014	0.1835 $\pm$ 0.0013	0.038	0.007
HPN-SteepDecay	0.1626 $\pm$ 0.0006	0.1602 $\pm$ 0.0010	0.1596 $\pm$ 0.0006	0.1486 $\pm$ 0.0006	0.1600 $\pm$ 0.0008	0.006	0.022

DGP outperforms IEC-only in all three environments ($p < 0.05$) and outperforms the capacity-matched PPO-GRU baseline in all three environments ($p < 0.03$). VCE-PPO does not outperform IEC-only in any environment (one-sided $p = 0.986, 0.999, 0.303$, H_1 : VCE-PPO $>$ IEC-only). DGP-Full is the weakest agent in all environments, confirming that the current EIL-style extension is harmful across all depletion regimes — this validates the NoEIL canonicalization.

The effect size (DGP – IEC-only) varies across regimes. Under slow recovery ($\rho=0.01$), depleted patches persist longer, making the viability threshold $\tau=0.25$ a noisier signal; DGP remains significantly better ($p=0.038$), confirming the mechanism is robust rather than tuned to a specific depletion regime. Under steep decay ($\delta=0.40$), rapid local collapse means τ is crossed after fewer exploits, producing more frequent pivot events; DGP maintains its advantage ($p=0.006$).

7. Mechanistic Diagnostics

7.1 Overcommit Index (OCI)

We define the Overcommit Index (OCI) as the fraction of EXPLOIT actions taken when local richness is below $\tau=0.25$ — a proxy for time wasted over-harvesting depleted cells.

Table 3. Overcommit Index (OCI) across agents and environments (mean $\pm$ SEM, 10 seeds).

Agent	HPN OCI	SlowRecovery OCI	SteepDecay OCI
DGP	0.202 $\pm$ 0.002	0.226 $\pm$ 0.002	0.391 $\pm$ 0.003
VCE-PPO	0.249 $\pm$ 0.005	0.269 $\pm$ 0.004	0.419 $\pm$ 0.003
DGP-Full	0.276 $\pm$ 0.007	0.305 $\pm$ 0.006	0.437 $\pm$ 0.003
IEC-only	0.346 $\pm$ 0.004	0.365 $\pm$ 0.003	0.522 $\pm$ 0.004
No-pivot†	0.422	—	—
PPO-GRU	0.320 $\pm$ 0.003	0.340 $\pm$ 0.006	0.481 $\pm$ 0.004

†No-pivot OCI was collected only on standard HPN in the main sweep; SlowRecovery and SteepDecay values are not available.

DGP achieves the lowest OCI in all environments. Notably, VCE-PPO reduces OCI relative to IEC-only (e.g., 0.249 vs 0.346 on HPN) — entropy modulation does change the agent’s behavior when depleted — but it does not close the gap with DGP (OCI gap DGP vs VCE-PPO: 0.047 HPN, 0.043 SlowRecovery, 0.028 SteepDecay, all $p < 0.001$). DGP-Full achieves lower OCI than IEC-only but higher OCI than DGP, indicating the EIL phase partially reduces overcommitment but introduces other costs that dominate performance. The OCI-AUC dissociation between VCE-PPO and DGP is the subject of Section 8.

Note on OCI measurement: OCI is defined using the same $\tau=0.25$ threshold that drives the viability gate. This is intentional — OCI measures behavior precisely in the low-viability regime where the gate fires — but it means OCI is architecturally aligned with the gate condition rather than being an independent behavioral measure. The key scientific value of OCI here comes from the dissociation: VCE-PPO is also exposed to the same threshold (its gate also fires at $\tau=0.25$) yet achieves higher OCI than DGP, confirming the measurement is capturing a real behavioral difference and not a construction artifact.

7.2 Pivot Trigger Alignment

Post-pivot reward (mean reward on the step immediately following a pivot event): **0.184**, compared to the overall mean reward of **0.209** across all steps. The gate fires on cells with richness below $\tau=0.25$ — cells that are genuinely poor or depleted — and the agent earns below-average reward on the pivot step itself, consistent with the gate firing on depleted cells. Recovery in subsequent steps (return to IEC on higher-richness cells) drives the AUC gain.

7.3 RESET_ALPHA Sensitivity

We ran a 3-seed diagnostic scan over $\alpha \in \{0.0, 0.1, 0.5, 1.0\}$ on standard HPN.

Table 4. RESET_ALPHA sensitivity diagnostic (3 seeds, standard HPN).

RESET_ALPHA	AUC (mean $\pm$ SEM, 3 seeds)	q4b/ep
0.0 (full wipe)	0.2071 $\pm$ 0.0008	93.1
0.1	0.2037 $\pm$ 0.0025	90.7
0.5 (canonical)	0.2082 $\pm$ 0.0008	95.6
1.0 (no reset)	0.2089 $\pm$ 0.0018	95.7
IEC-only	0.2040 $\pm$ 0.0009	—

All pivot variants outperform IEC-only in this diagnostic; differences across α values are not statistically distinguishable at $n = 3$. Notably, $\alpha = 1.0$ (no memory wipe, pure temperature switch) achieves the highest diagnostic mean (0.2089), suggesting the high-temperature reorientation signal is a primary contributor to the pivot benefit. Memory disruption ($\alpha < 1.0$) provides additional robustness but is not strictly necessary for the benefit to appear. We adopt $\alpha = 0.5$ as the canonical configuration, consistent with the 10-seed main result (DGP: 0.2086 $\pm$ 0.0007).

8. VCE-PPO Control: Entropy Modulation vs Structured Pivot Action

To test whether the DGP benefit reduces to viability-conditioned entropy modulation, we ran a pre-registered comparison against **Viability-Conditioned Entropy PPO (VCE-PPO)**. VCE-PPO uses the same GRU backbone and the same hard step-gate at $\tau=0.25$, but instead of executing a structured pivot action, it increases its policy entropy coefficient when

$V_t < \tau$ (entropy_high selected by calibration; see §8.2) and reverts to low entropy (0.01) otherwise. The critical distinction is not the presence of a viability gate: both DGP and VCE-PPO use an identical hard step-gate at the same threshold. The difference is that DGP couples the gate to a structured spatial pivot action, whereas VCE-PPO couples the gate only to increased policy entropy.

8.1 Pre-Registered Decision Rule

Branch classification was specified before running:

- **Branch A:** VCE-PPO OCI > DGP OCI by ≥ 0.02 absolute, $p < 0.05$, AND DGP AUC not worse by $> 0.002 \rightarrow$ structured pivot is the key performance-improving mechanism in this environment family
- **Branch B:** differences within that window
- **Branch C:** DGP AUC worse by ≥ 0.003 OR OCI not improved and overall favors VCE-PPO

8.2 Entropy Schedule Calibration

A preliminary calibration sweep (3 seeds $\times$ 3 entropy values: 0.10, 0.15, 0.20) on standard HPN identified entropy_high=0.10 as the strongest VCE-PPO variant by AUC (0.1985 vs 0.1972 for 0.15 and 0.1950 for 0.20). Higher entropy values reduced OCI further (VCE-020 OCI=0.221) at cost of AUC. To ensure the strongest possible VCE-PPO variant competed against DGP, VCE-010 (entropy_high=0.10) was selected and the entropy schedule was frozen before the full 10-seed comparison was run. No further tuning of the entropy schedule was performed after seeing the full results.

8.3 Results

Table 5. VCE-PPO comparison results (mean $\pm$ SEM, 10 seeds).

Environment	Agent	AUC mean $\pm$ SEM	OCI mean $\pm$ SEM
HPN	DGP	0.2086 $\pm$ 0.0007	0.2019 $\pm$ 0.0024
HPN	VCE-PPO	0.1999 $\pm$ 0.0013	0.2487 $\pm$ 0.0047
HPN	IEC-only	0.2038 $\pm$ 0.0005	0.3459 $\pm$ 0.0035
SlowRecovery	DGP	0.1881 $\pm$ 0.0007	0.2262 $\pm$ 0.0024
SlowRecovery	VCE-PPO	0.1811 $\pm$ 0.0009	0.2691 $\pm$ 0.0035
SlowRecovery	IEC-only	0.1854 $\pm$ 0.0008	0.3649 $\pm$ 0.0034
SteepDecay	DGP	0.1626 $\pm$ 0.0006	0.3910 $\pm$ 0.0035

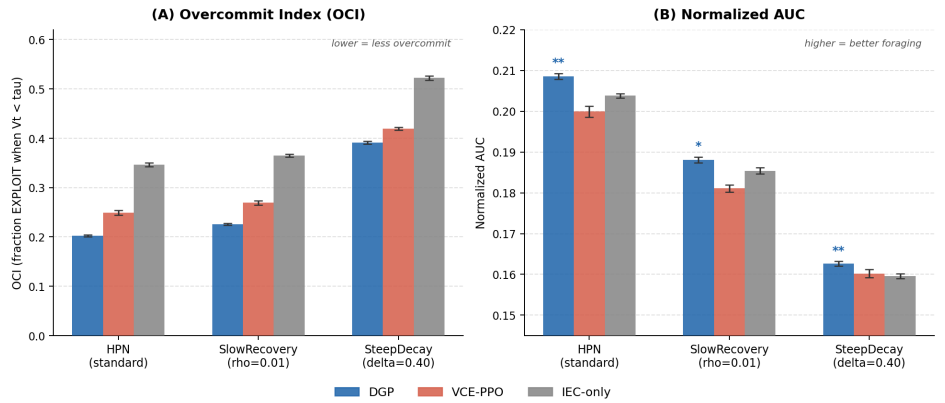
Environment	Agent	AUC mean $\pm$ SEM	OCI mean $\pm$ SEM
SteepDecay	VCE-PPO	0.1602 $\pm$ 0.0010	0.4193 $\pm$ 0.0035
SteepDecay	IEC-only	0.1596 $\pm$ 0.0006	0.5225 $\pm$ 0.0042

Table 6. Branch classification summary.

Environment	OCI gap (VCE-DGP)	p (OCI)	AUC gap (DGP-VCE)	Branch
HPN	0.047	< 0.001	+0.0087	A
SlowRecovery	0.043	0.001	+0.0071	A
SteepDecay	0.028	< 0.001	+0.0024	A

Branch A is confirmed in all three environments. All OCI gaps exceed the pre-registered 0.02 threshold and are statistically significant. DGP’s AUC advantage is positive in all cases, well above the -0.002 floor.

Note on SteepDecay: The Branch A margins are tightest in this most compressed depletion regime (OCI gap 0.028, AUC gap 0.0024). SteepDecay confirms the pattern holds under extreme depletion stress, but HPN and SlowRecovery carry the main result with larger margins.



VCE-PPO reduces OCI vs IEC-only but fails to improve AUC ($p > 0.30$). DGP reduces OCI further and improves AUC (** $p < 0.01$, * $p < 0.05$ vs IEC-only). Error bars = ± 1 SEM, $n=10$ seeds.

Figure 3

Figure 3. Key dissociation result across three environments. **(A)** Overcommit Index (OCI): VCE-PPO reduces OCI relative to IEC-only, but DGP reduces it substantially more. **(B)** Normalized AUC: VCE-PPO fails to outperform IEC-only in any environment (one-sided $p > 0.30$), while DGP significantly outperforms IEC-only (** $p < 0.01$, * $p < 0.05$). The dissociation — OCI reduction without AUC gain for VCE-PPO; OCI reduction with AUC gain for DGP — identifies the structured pivot action

as the load-bearing mechanism in this environment family. Error bars = ± 1 SEM, $n = 10$ seeds.

8.4 Mechanistic Interpretation

Two dissociations identify the mechanism:

VCE-PPO reduces OCI relative to IEC-only but does not improve AUC. VCE-PPO achieves lower OCI than IEC-only in all environments (0.249 vs 0.346 on HPN), confirming that entropy increase does change the agent's behavior during low-viability periods. However, this OCI reduction does not translate into AUC gains: VCE-PPO fails to outperform IEC-only on AUC in any environment (one-sided $p = 0.986, 0.999, 0.303$ for HPN, SlowRecovery, SteepDecay; H_1 : VCE-PPO > IEC-only). Entropy modulation causes the agent to explore more diffusely when depleted, but without a structured spatial pivot action, it continues to explore near the depleted patch rather than escaping to richer areas.

DGP reduces OCI further and converts that reduction into AUC gains. DGP achieves substantially lower OCI than VCE-PPO (0.202 vs 0.249 on HPN; 0.391 vs 0.419 on SteepDecay) and significantly outperforms IEC-only on AUC across all environments ($p = 0.001, 0.038, 0.006$). The structured pivot action — a single high-temperature step that disrupts the current exploitation trajectory and drives spatial relocation — is what converts viability-conditioned behavior change into actual return gains in this environment family.

The conclusion is that the viability gate alone is not sufficient: entropy adjustment produces diffuse local re-exploration, but spatial escape — physically relocating under the high-temperature pivot signal — is the key performance-improving step in this environment family.

9. Scope Limitation: MPB

DGP fails on the Multi-Patch Bandit (0.129 ± 0.011 vs IEC-only 0.196 ± 0.002 ; 3-seed diagnostic, Appendix C; $p \approx 1.0$ for DGP > IEC-only). The mechanism requires spatial structure: in HPN, the pivot disrupts memory and the agent moves spatially, escaping depleted cells. In MPB (no spatial structure), pivot-then-harvest cycles the agent back to the same depleted patch ($q4b=216/ep$ vs $94/ep$ in HPN).

Forced-switch diagnostic (Appendix C): We tested a variant (DGP-ForcedSwitch) that masks the HARVEST logit immediately post-pivot, forcing the agent to select a SWITCH action. This eliminates the pivot-then-harvest loop but does not help: ForcedSwitch AUC = 0.065 ± 0.0003 — dramatically below Standard DGP (0.129) and IEC-only (0.196), with $q4b=177/ep$. The interpretation: forcing a switch moves the agent to a different depleted patch, which immediately triggers another pivot, producing a destructive switch-pivot cycle. This confirms the scope limitation is about spatial escape — physical relocation in a spatial environment — not about which action follows pivot. The depletion gate cannot substitute for movement in non-spatial settings.

10. Discussion

What DGP does: The depletion gate converts a viability-drop signal into a reorientation event: one high-temperature action selection step that disrupts the current exploitation trajectory and drives spatial relocation. The mechanism is effective under rapid patch depletion (SteepDecay: pivot after ~ 1 exploit) and remains beneficial under slower dynamics. The VCE-PPO control establishes a clean causal account: entropy modulation alone is not sufficient; the structured pivot action — physical relocation under high-temperature selection — is the key performance-improving step in this environment family.

The hard-gate vs entropy-modulation distinction. The contrast between DGP and VCE-PPO is not hard viability-gating versus smooth or learned entropy modulation. Both agents use an identical hard step-gate at $\tau=0.25$; neither uses a learned continuous entropy schedule. The mechanistic difference is that DGP couples the gate to a structured spatial pivot action, while VCE-PPO couples it only to increased policy entropy. This precision matters for interpretation: the result does not show that rule-based gating outperforms learned gating, but that the pivot action is the load-bearing component of the gate-conditioned behavioral change in this environment family.

What EIL was supposed to do and why it failed: The EIL drift phase was designed to provide a “concentrated exploitation” mode between IEC cycles. In practice, brief EIL bursts (mean 3.3 steps) provide insufficient time for meaningful exploitation learning and the slow memory drift adds noise. DGP-Full (IEC + EIL + Pivot) underperforms DGP in all three

environments (AUC 0.189 vs 0.209 on HPN; 0.174 vs 0.188 on SlowRecovery; 0.149 vs 0.163 on SteepDecay), confirming that the EIL phase is harmful regardless of whether a depletion gate is present. This confirmatory negative result validates the NoEIL canonicalization and suggests that EIL-style drift phases as currently implemented are not a productive direction for this environment family.

Scope and limitations: DGP is a spatial foraging mechanism. The pivot disruption drives reorientation only when coupled to physical movement — the necessary coupling in this environment family. The forced-switch diagnostic (Appendix C) rules out the alternative explanation that the scope limitation is purely about action selection: even when post-pivot harvesting is blocked, the agent enters a switch-pivot cycle on the same depleted patch family. Future work could design a bandit-compatible variant that maintains an explicit patch quality map and selects the highest-richness arm post-pivot.

Claim scope. These results establish the structured pivot action as the key performance-improving mechanism in this environment family (HPN and variants). The result does not constitute a general claim that depletion-gated pivoting is necessary or sufficient across all foraging settings. Broader generalization — to non-spatial settings, partial observability, stochastic depletion, or larger maps — requires further investigation.

Biological interpretation: The IEC→PIVOT→IEC cycle mirrors the give-up time heuristic in optimal foraging: IEC is committed exploitation, PIVOT is triggered departure upon sufficient depletion, re-entering IEC represents starting fresh on a new patch.

10.1 Limitations

Training architecture. The GRU agent is trained with detached hidden states per step (no BPTT across timesteps within an episode). This is a deliberate simplification targeting the minimal mechanism question: we ask whether a simple depletion-gated rule can improve foraging, not whether a fully-trained recurrent agent with long-horizon credit assignment would do so. The answer is yes in these environments. Future work should test whether full BPTT changes the comparative picture — particularly whether PPO-GRU with full recurrent training narrows or eliminates the DGP advantage.

Environment scope. All environments in this study are variants of a custom 7×7 grid family (HPN). While the three variants span meaningfully different depletion dynamics, they share the same basic spatial structure, observation design, and action space. The mechanism result holds within this family; whether it extends to standard foraging benchmarks (e.g., MiniGrid, Crafter-derived tasks, larger maps with partial observability) is an open question and the primary target for follow-up work.

EIL scope. The claim that the EIL-style drift phase is harmful is specific to the current implementation (brief drift bursts of mean 3.3 steps, TEMP_EIL=0.7) and this environment family. We do not claim to have falsified all possible exploitation-phase architectures. A properly designed exploitation mode with longer commitment windows or learned termination conditions may behave differently.

Viability signal. $V_t = \text{raw local richness}(\text{obs}[2])$ is a simple proxy that fires on both genuinely depleted cells and inherently poor cells. A more precise depletion detector (e.g., a learned depletion rate signal) could in principle improve gate timing, but testing this is outside the scope of the current mechanism study.

Acknowledgments

The broader lifecycle framework motivating this work was originally inspired in part by observations from the Objective Personality System developed by Dave Powers and Shan Renée. Their work is cited here solely as attribution of inspiration; all empirical claims, test designs, and conclusions rest exclusively on the experiments reported in this manuscript.

Declaration of Generative AI and AI-Assisted Technologies

During the preparation of this work the author used Claude (Anthropic) to assist with code implementation, editing clarity, and consistency checking. After using this tool, the author reviewed and edited the content as needed and takes full responsibility for the content of the publication.

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Appendix

A. Experimental Details

Environment hyperparameters:

Parameter	HPN Standard	HPN-SlowRecovery	HPN-SteepDecay	MPB	SNW
Grid / Patches	7×7	7×7	7×7	10 patches	7×7
EXPLOIT_DECAY (δ)	0.60	0.60	0.40	0.70	—
RECOVERY_RATE (ρ)	0.03	0.01	0.03	0.02	—
RECOVERY_WINDOW	50 steps	50 steps	50 steps	—	—
Initial richness	Exp(0.7) clipped [0,1]	same	same	Exp(0.7)	1.0 (fixed)
Max steps/episode	400	400	400	500	400
Observation dim	6	6	6	6	5

Agent hyperparameters (all agents):

Parameter	Value
GRU hidden dim	64
Learning rate	3e-4
Entropy coefficient	0.01
Discount γ	1.0 (episodic)
Optimizer	Adam
Training episodes	500
Seeds	10 (main), 3 (diagnostics)
BPTT	None (detached hidden state per step)

DGP-specific hyperparameters:

Parameter	Value
T_IEC (IEC temperature)	2.0
T_PIVOT (pivot temperature)	5.0
TAU (viability threshold)	0.25
RESET_ALPHA (α , canonical)	0.5
V_t signal	raw local richness (obs[2])

B. RESET_ALPHA Scan

3-seed diagnostic on standard HPN. All agents use the same GRU backbone as DGP canonical.

RESET_ALPHA	s0	s1	s2	mean	SEM
0.0	0.2063	0.2062	0.2088	0.2071	0.0008
0.1	0.2051	0.1989	0.2071	0.2037	0.0025
0.5	0.2093	0.2067	0.2088	0.2082	0.0008
1.0	0.2116	0.2098	0.2054	0.2089	0.0018
IEC-only	0.2053	0.2044	0.2022	0.2040	0.0009

C. MPB Forced-Switch Diagnostic

3-seed diagnostic. DGP-ForcedSwitch masks HARVEST logit for one step post-pivot.

Agent	s0	s1	s2	mean	SEM	q4b/ep
ForcedSwitch	0.0651	0.0640	0.0648	0.0646	0.0003	176.7
Standard DGP	0.1268	0.1114	0.1482	0.1288	0.0107	215.5
IEC-only	0.1986	0.1936	0.1970	0.1964	0.0015	—

ForcedSwitch is markedly worse than both Standard DGP and IEC-only, confirming that the scope limitation is spatial, not addressable by action masking.

D. Full Per-Seed Results

Full per-seed AUC and OCI data are available in the supplementary data files: HPN_SWEEP.json (main 10-seed sweep), HPN_NOEIL.json (NoEIL ablation), and HPN_VARIANTS.json (SlowRecovery and SteepDecay variants). VCE-PPO full per-seed results are provided in Appendix E.

E. VCE-PPO Full Per-Seed Results

Source: VCE_SWEEP.json, 10 seeds $\times$ 3 environments.

HPN — AUC per seed:

Agent	s0	s1	s2	s3	s4	s5	s6	s7	s8	s9	mean
DGP	0.2093	0.2067	0.2088	0.2074	0.2103	0.2116	0.2086	0.2044	0.2069	0.2122	0.2086
VCE-PPO	0.1983	0.2028	0.1942	0.2028	0.1946	0.1982	0.1966	0.2013	0.2027	0.2075	0.1999
IEC-only	0.2053	0.2044	0.2022	0.2034	0.2007	0.2056	0.2069	0.2035	0.2029	0.2033	0.2038
DGP-Full	0.1936	0.1893	0.1804	0.1931	0.1907	0.1971	0.1780	0.1856	0.1860	0.1949	0.1889
PPO-GRU	0.2001	0.2034	0.2003	0.2066	0.2004	0.2044	0.1987	0.2034	0.2024	0.2044	0.2024

HPN — OCI per seed:

Agent	s0	s1	s2	s3	s4	s5	s6	s7	s8	s9	mean
DGP	0.2167	0.2092	0.2015	0.2117	0.1968	0.1967	0.2004	0.1943	0.1969	0.1949	0.2019
VCE-PPO	0.2638	0.2503	0.2711	0.2241	0.2612	0.2467	0.2615	0.2382	0.2389	0.2308	0.2487
IEC-only	0.3646	0.3360	0.3509	0.3516	0.3540	0.3384	0.3242	0.3539	0.3369	0.3482	0.3459
DGP-Full	0.2963	0.2800	0.2500	0.2917	0.2926	0.2913	0.2401	0.2911	0.2367	0.2897	0.2760
PPO-GRU	0.3358	0.3166	0.3261	0.3071	0.3231	0.3265	0.3241	0.3147	0.3060	0.3225	0.3202